

Nicole Sorensen

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Education

Carnegie Mellon University, Pittsburgh, PA

Expected May 2026

B.S. in Statistics and Machine Learning

GPA: 3.88, Dean's List

Relevant Coursework: Causation and Machine Learning, Intro to Causal Inference, Intro to Machine Learning, Machine Learning in Practice, Probability and Statistical Inference, Modern Regression (Fall 2025), Computational Methods for Biological Modeling and Simulation (Fall 2025), Data Structures and Algorithms, Multivariate Analysis

Research Experience

Causal Single-Cell Differential Expression Analysis, Carnegie Mellon University 08/2025 – Present

- Evaluating CSDE (Causal Single-cell Differential Expression) method against baseline approaches on Alzheimer's disease single-cell RNA-seq data from microglia populations (SEAAD MTG dataset).
- Conducted enrichment analysis using Fisher's exact test and GSEA to assess overlap with known AD-associated genes from Open Targets, and performed LD score regression to evaluate SNP heritability enrichment near CSDE-identified genes.
- Implemented donor-balanced sampling and pseudobulk aggregation pipelines in Python (scanpy, pandas) to control for batch effects and demographic confounders including APOE genotype.

Bridges to Healthcare Technology Program, CMU, Sponsored by Optum

06/2024 – 08/2024

- Investigated socioeconomic drivers of racial disparities in hospitalizations using County Health Rankings and Roadmaps data.
- Applied generalized additive models (GAMs) and random forests (R, scikit-learn) to identify predictive variables such as income and insurance status. Presented findings to CMU faculty and UnitedHealth Group professionals.

Professional Experience

Optum (UnitedHealth Group), Data Science Intern

05/2025 – 08/2025

- Conducted root cause analysis and pattern identification using LLM pipelines, k-means clustering, topic modeling, and SQL analytics (Databricks, Snowflake) on consumer and tech ops data.

Teaching Assistant, Probability and Statistical Inference I, CMU

08/2025 – Present

- Lead recitations, held office hours, and graded for undergraduate statistics course.

Projects

Causal Analysis of ICU Transfers and Sepsis, CMU

11/2024 – 05/2025

- Investigated the causal impact of ICU transfers on sepsis risk using MIMIC-IV data
- Applied the Fast Causal Inference (FCI) algorithm with kernel-based conditional independence tests (KCI) to identify potential mediators such as Glasgow Coma Scale and creatinine.
- Estimated a negative average treatment effect ($ATE = -0.174$), suggesting ICU transfer may reduce sepsis risk.

Temporal Prediction of Ventilation using Deep Learning, CMU

10/2024 – Present

- Developed a multi-input deep learning model (LSTM + ClinicalBERT) to predict ICU ventilation needs 6 hours in advance using MIMIC-IV, enabling earlier interventions
- Achieved 7× improvement over baseline in PR AUC by applying Boruta feature selection and oversampling to handle class imbalance

Technical Skills

- **Statistical Genetics & Genomics:** Single-cell RNA-seq analysis (scanpy), LD score regression, enrichment analysis (GSEA, Fisher's exact test), differential expression, GWAS methods
- **Causal Inference:** Causal discovery (FCI, KCI), treatment effect estimation (ATE/ITE), propensity scoring, confounding adjustment
- **Machine Learning & Statistics:** Deep learning (PyTorch, TensorFlow, LSTM), random forests, GAMs, feature selection, clustering, mixed-effects models
- **Programming:** Python (pandas, NumPy, scikit-learn, statsmodels), R (tidyverse, ggplot2), SQL, C — Databricks, Snowflake, Azure AI